

UniCRM System Architecture

Multi-Layered Architecture



Domain Layer

Pure business entities and core data structures. Defines the fundamental relationships between Users, Courses, and Research Papers without external dependencies.



Service Layer

Encapsulates complex business logic. Orchestrates interactions between repositories and domain models to handle grading, registration, and reporting.



Persistence Layer

Abstracted data storage using the Repository pattern. Handles serialization to JSON/DAT formats while maintaining a consistent interface for the application.

Core Functional Modules

Academic & Comm

Management: Handles student enrollment, course offerings, and room assignments with capacity constraints.

Support: Integrated ticketing system for tech requests and a secure messaging framework for staff communication.

Research Framework

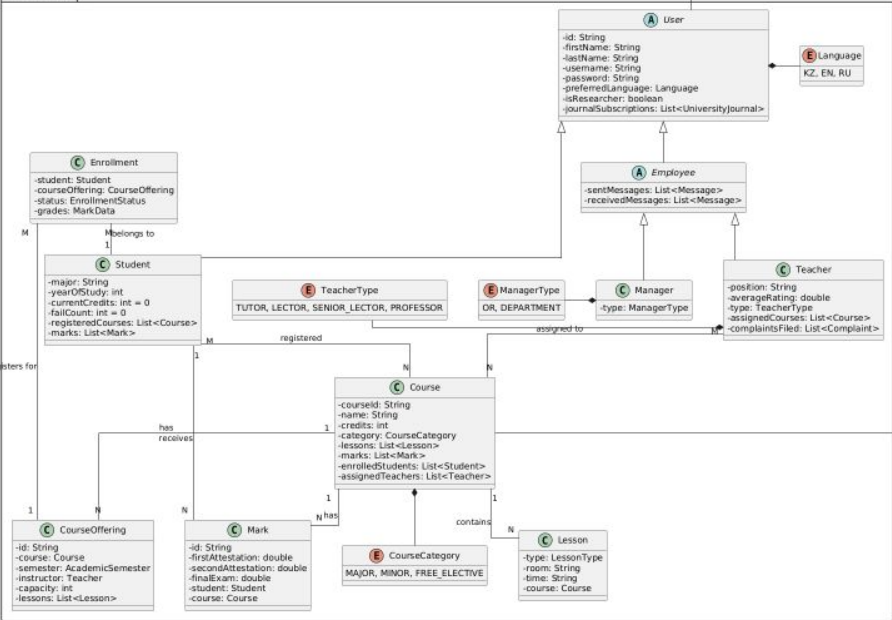
Scholarship: Advanced module for publishing papers, managing university journals, and automating citations.

Metrics: Dynamic calculation of academic impact, including H-Index tracking and researcher performance analysis.

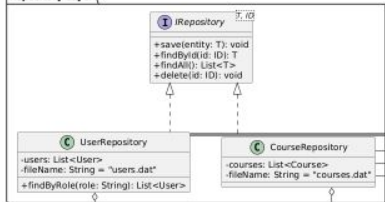
Engineering Excellence

-  **Command Pattern** Decouples user interactions from business logic, enabling a flexible, history-aware, and extensible CLI interface.
-  **Decorator Pattern** Dynamically enhances base User objects with Research capabilities, avoiding rigid inheritance hierarchies.
-  **Repository Pattern** Provides a clean abstraction over file-based storage, allowing for future migrations to SQL databases with zero logic changes.
-  **Singleton & State Management** Centralized UserSession management ensures secure authentication and consistent global access to the current system state.

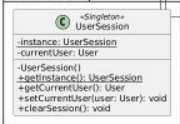
Domain Layer



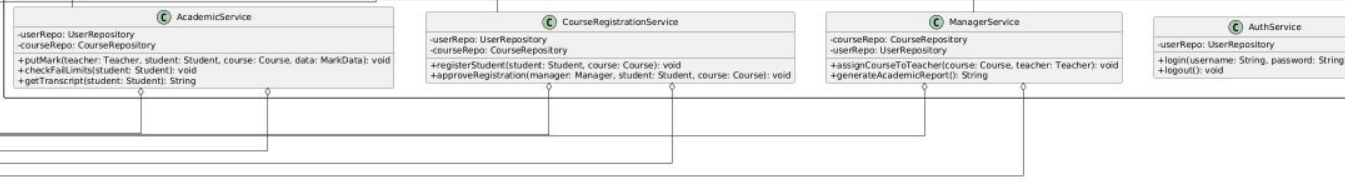
Repository Layer



Session & State



Service Layer



Command Layer

